

RC CIRCUIT

Sailesh Mishra

BASICS . . .

capacitance = $10 \mu\text{F}$
 $= 10 \times 10^{-6} \text{ F}$
 $V = 5 \text{ V}$
 $Q = C \cdot V$
 $= 10 \mu\text{F} \times 5$
 $= 50 \mu\text{C}$

$V = \frac{Q}{C}$

$Q = ne$

$\frac{50 \mu\text{C}}{1.6 \times 10^{-19}} = n_e = 3.125 \times 10^{14}$
 $= 300 \text{ no. of } e^-$

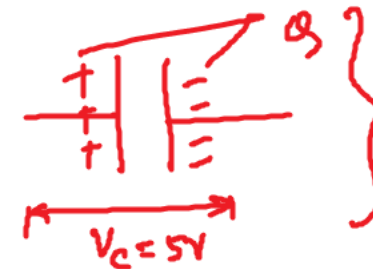
1SD +ve $\rightarrow$ +ve METAL PLATE.
 1SD -ve $\rightarrow$ -ve METAL PLATE.

$i = \frac{V}{R}$

Similarly;

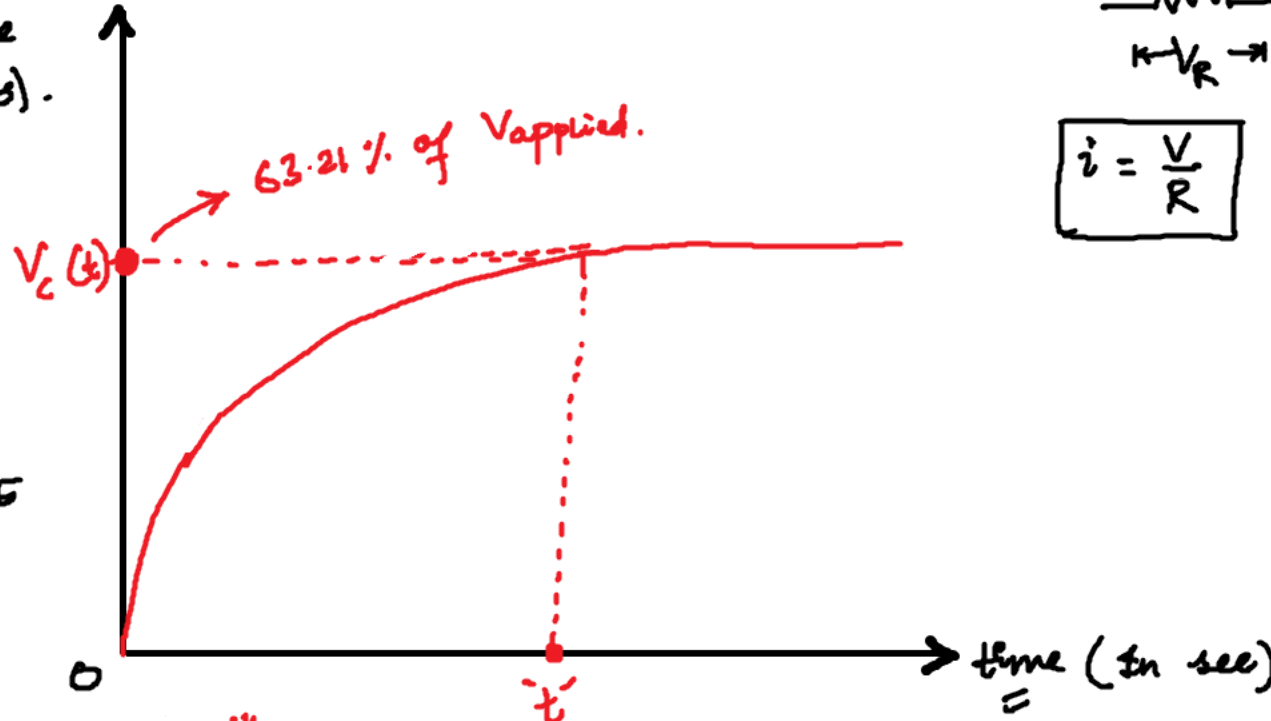
ex. SiO_2 : VL61
 $K = 3.97 \approx 4$

$\text{Al}_2\text{O}_3 = 9.0 \text{ (v. high)}$
 $\text{Air} = 1.0 \text{ (K)}.$
 $\text{H}_2\text{O}_2 = 20-25.$



$Q = CV$

V or Battery is always const.



V
(mV)
volts.

$V_C(t)$: CHARGING $\rightarrow V_C(t) = V_{app} (1 - e^{-t/\tau})$
 $\downarrow$
 $1.25V = 1.8V (1 - e^{-t/\tau})$



V_{app}

1.8V

$i' \rightarrow 1.8V$

$$\begin{cases} \tau = R \cdot C \\ = 1k\Omega \times 10\mu F \\ = 1 \times 10^3 \times 10 \times 10^{-6} F. \end{cases}$$

$V_C(t)$: DISCHARGING $\rightarrow V_C(t) = V_{app} \times (e^{-t/\tau})$

1.2V $\rightarrow$ 1.25V

0.9

0.45

$i' \rightarrow 0V$

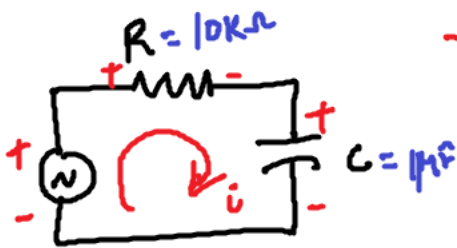
t_n

t (in ms)

Example-1 :-



$V_{\text{pulse}} = 1.8\text{V}$



$$\rightarrow V_{\text{pulse}} - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_R - V_C = 0$$

$\Rightarrow 1.8 - V_C = V_R$: voltage drop across $R = 10\text{k}\Omega = 10 \times 10^3 \Omega$.
voltage drop across $C = 1\mu\text{F} = 1 \times 10^{-6} \text{F}$.

$$P = \frac{V^2}{R} + \frac{1}{2} CV^2$$

Time period = 8ms

$T_{\text{ON}} = 4\text{ms}$

$T_{\text{rise}} = T_{\text{fall}} = 0\text{s}$

Q1 Find $i \rightarrow R$

Q2 Find $i \rightarrow C$

$$\tau = R \cdot C = 10\text{k}\Omega \times 1\mu\text{F} = 10 \times 10^3 \Omega \times 10^{-6} \text{F} = 10^{-2} \text{s} = 10\text{ms}$$

$$\rightarrow i_R(t) = \frac{V}{R} = \frac{V_N}{R} = \frac{V_{\text{pulse}} - V_C}{R} = \frac{1.8\text{V} - V_{\text{app}} \times (1 - e^{-t/\tau})}{R}$$

$$\rightarrow i_C(t) = C \cdot \frac{dV}{dt} = \frac{1\mu\text{F} \times V_{\text{app}} \times (1 - e^{-t/\tau})}{t}$$

meet.google.com is sharing your screen.

Stop sharing

Hide

1920 x 930px

100%

100%

100%

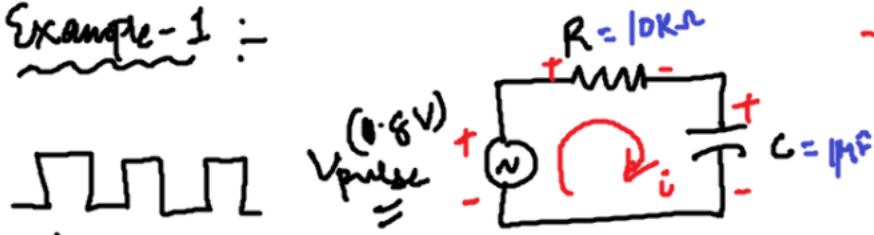
100%

100%

100%

100%

Example-1 :



Time period = 8ms

$T_{ON} = 4ms$

$T_{rise} = T_{fall} = 0s$

Q1: Find $V_C(t)$ at $2ms$; when CAPACITOR is charging...

Q2: Find $V_C(t)$ at $5ms$; when CAPACITOR is discharging...

$$\rightarrow V_{pulse} - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_C = V_R$$

→ voltage drop across $R = 10k\Omega = 10 \times 10^3 \Omega$.

→ voltage drop across $C = 1\mu F = 1 \times 10^{-6} F$.

$$\rightarrow \tau = R \cdot C = 10k\Omega \times 1\mu F = 10 \times 10^3 \Omega \times 10^{-6} F = 10^{-2} = 10^{-2}$$

$$= \frac{1}{100} = 10ms$$

$$\rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau})$$

$$\rightarrow V_C(t) = V_{app} \times e^{-t/\tau}$$

t_i : @ 0ms $\rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-0/10 \times 10^{-3}}) = 1.8 \times (1 - 1) = 0V$ ✓

t_i : @ 1ms $\rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-1ms/10ms}) = 1.8 \times (1 - e^{-0.1}) = 0.171V$ ✓

t_i : @ 2ms $\rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-2ms/10ms}) = 1.8 \times (1 - e^{-0.2}) = 0.326V$ ✓

$$t: @ 0ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-0}) = 1.8 \times (1 - 1) = \boxed{0V}$$

$$t: @ 1ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-\frac{1ms}{10ms}}) = 1.8 (1 - e^{-0.1}) = \boxed{0.171V} \checkmark$$

$$t: @ 2ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-\frac{2ms}{10ms}}) = 1.8 (1 - e^{-0.2}) = \boxed{0.326V} \checkmark$$

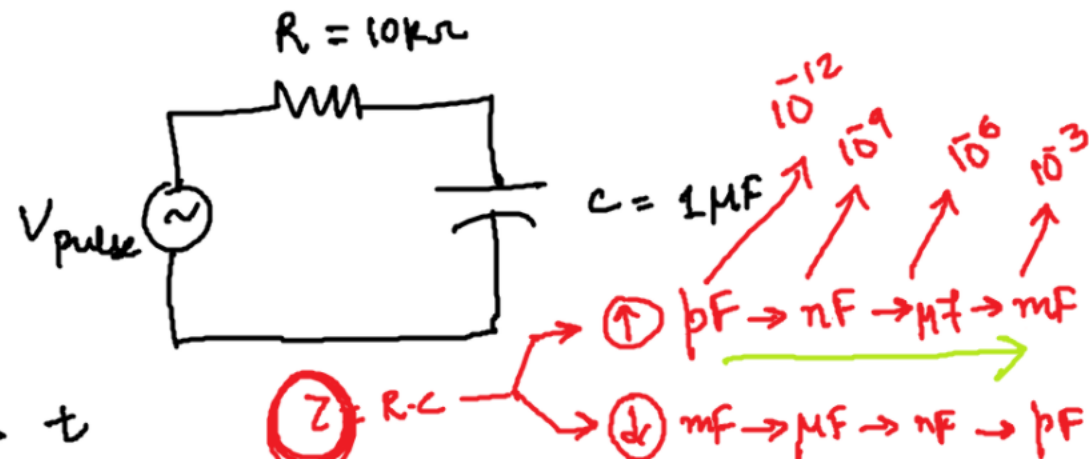
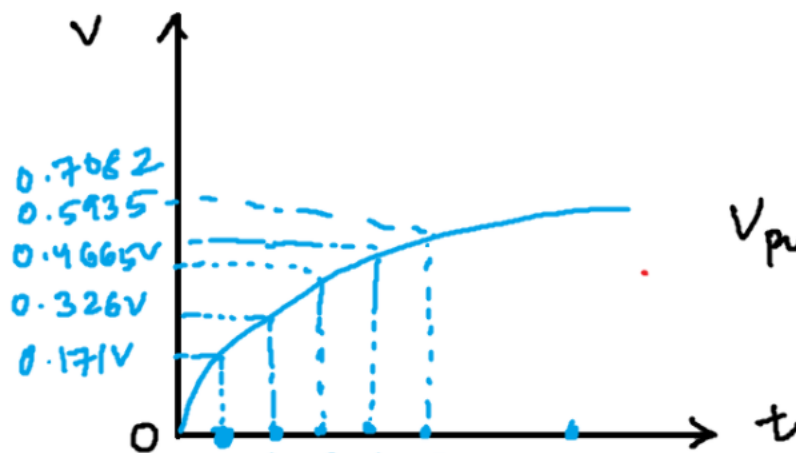
$$t: @ 3ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8 (1 - e^{-0.3}) = \boxed{0.4665V} \checkmark$$

$$t: @ 4ms \rightarrow V_c(t) = 1.8 (1 - e^{-0.4}) = \boxed{0.5934V}$$

$$t: @ 5ms \rightarrow V_c(t) = 1.8 (1 - e^{-0.5}) = \boxed{0.7082V} \rightarrow 6ms = \boxed{0.8121V} \rightarrow 7ms = \boxed{0.9041V} \rightarrow 8ms = \boxed{0.9912V}$$

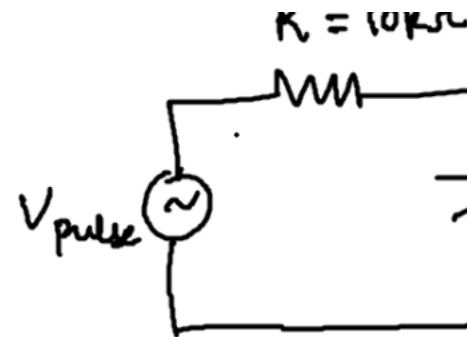
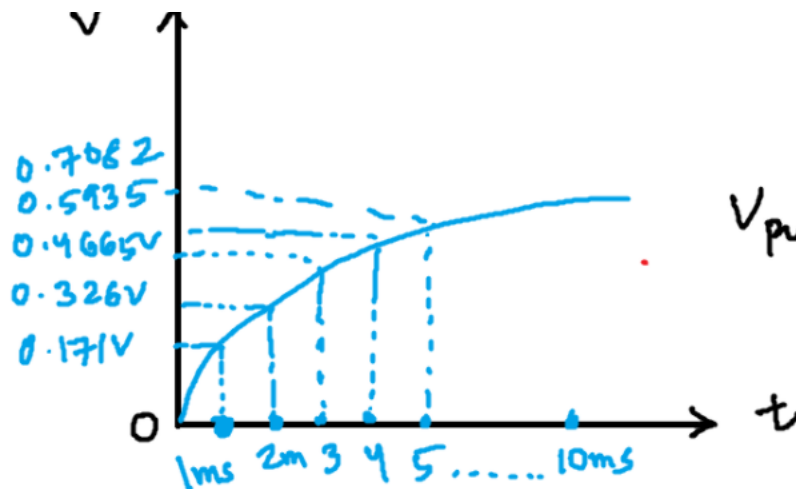
$\uparrow$ → FASTER charging
discharging

$\downarrow$ → SLOWER charging
discharging



$Z \uparrow \rightarrow$ FASTER charging discharging

$Z \downarrow \rightarrow$ SLOWER charging discharging



$C = 1\mu F$

$\uparrow$ pF $\rightarrow$ nF $\rightarrow$ μF $\rightarrow$ mF

$\downarrow$ mF $\rightarrow$ μF $\rightarrow$ nF $\rightarrow$ pF

(10^3) (10^{12})

10^{-12} 10^{-9} 10^{-6} 10^{-3}

$$V_C(t) = V_{\text{pulse}} \times (1 - e^{-t/\tau})$$

$$= 1.8V (1 - e^{-\frac{1\mu s}{10\mu s}}) = 1.8 \times (1 - e^{-0.1})$$

$$= \boxed{0.171V}$$

V

V_{applied}



@ 2ms (t)

$$V(t) = 1.8 (1 - e^{-0.2})$$

discharging.

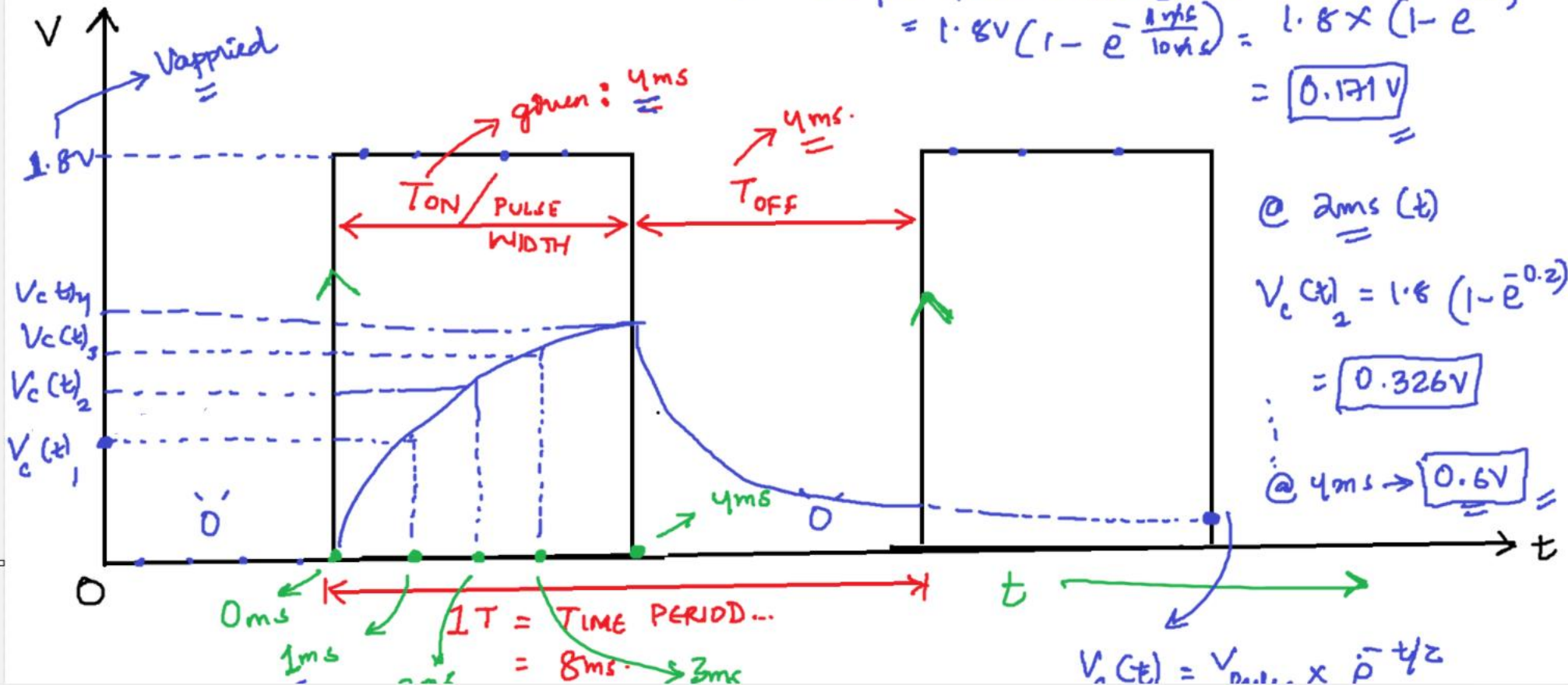
0 1ms 2ms 3 4 5..... 10ms

(10⁻³) (10⁻²)

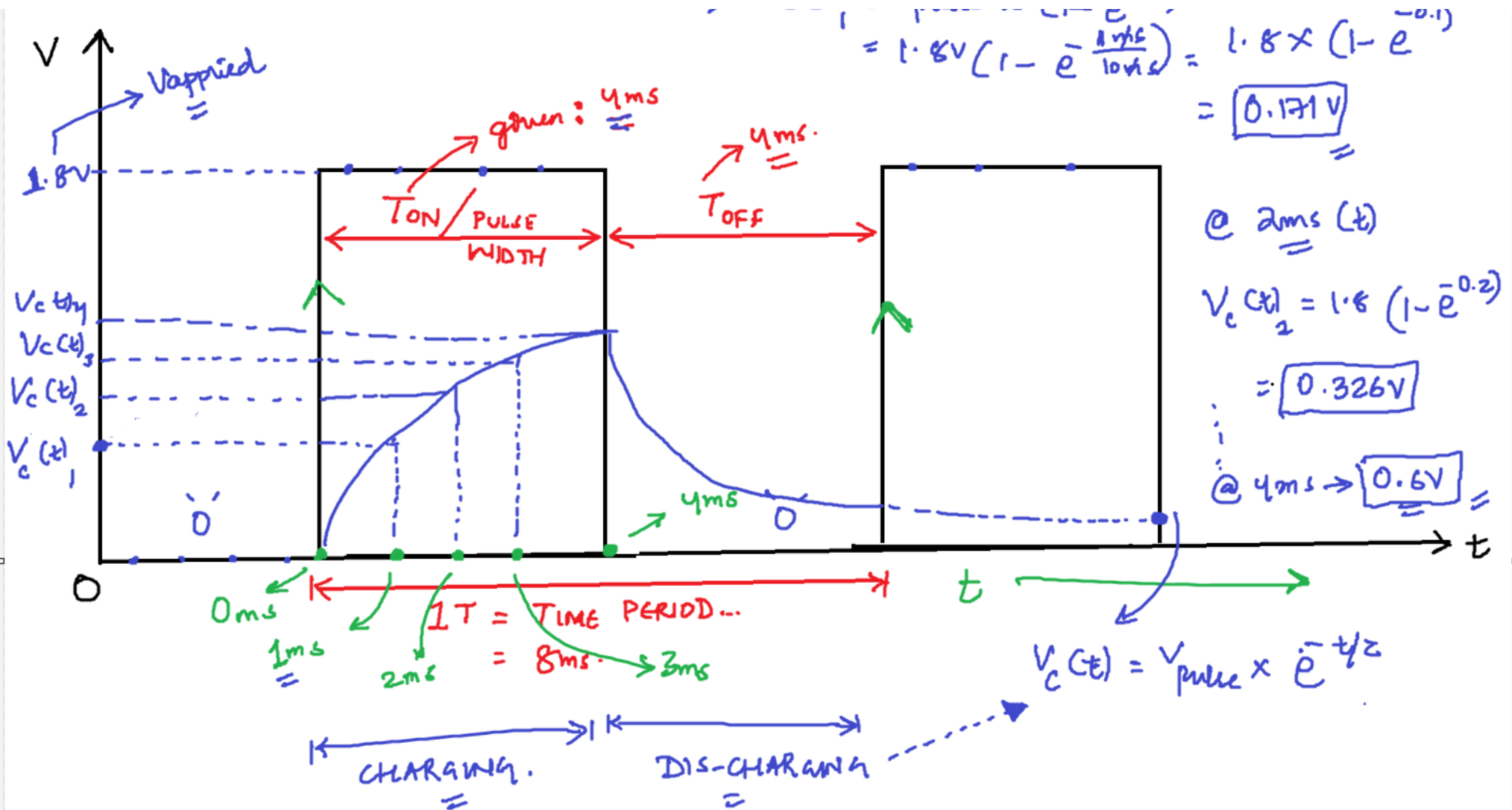
$$V_c(t)_1 = V_{\text{pulse}} \times (1 - e^{-t/\tau})$$

$$= 1.8V \left(1 - e^{-\frac{4\text{ms}}{10\text{ms}}}\right) = 1.8 \times (1 - e^{-0.4})$$

$$= \boxed{0.171V}$$



79%



CONCLUSION $\rightarrow \therefore T_{ON} = 4ms$

But @ 4ms ; Cap $\rightarrow 0.6V$ i.e. $\frac{1}{3}^{rd}$ of the Applied Voltage. (1.8V)

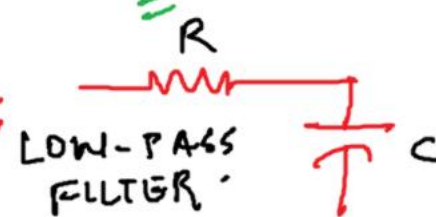
$\rightarrow$ Hence; it Needs Enough time to get CHARGED to its FULL POTENTIAL.

$\hookrightarrow$ This is NOTHING BUT DELAY !!

S_o; R-C ckt produces delays to the ckt.

$\boxed{*}$ ALSO ACTS AS A "FILTER"
 $\downarrow$
SS, DSP...

$R \rightarrow C$;



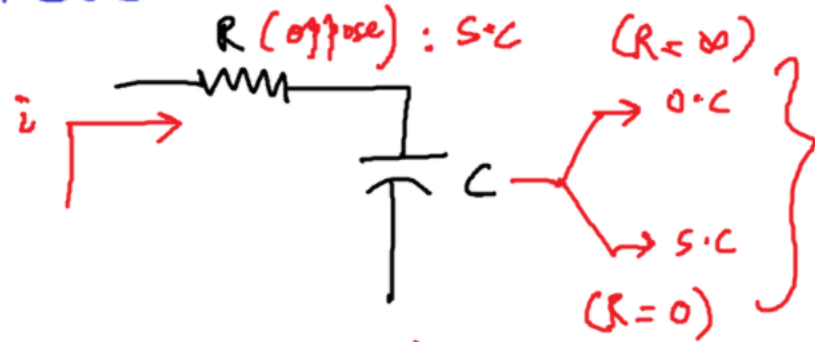
$C \rightarrow R$



CONCEPT OF LOW-PASS FILTER :-

R (oppose) : $S \propto C$ ($R < \infty$)

CONCEPT OF LOW-PASS FILTER :-



$t \rightarrow$
 $f = \frac{1}{t}$ ①



$\bar{R}' \rightarrow \infty \rightarrow O.C$
 $\bar{R}' \rightarrow 0 \rightarrow S.C$

PASS
FILTER.

$\rightarrow$ impedance.
 $R_c = X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

- D.C $\rightarrow f = 0 \text{ Hz} \rightarrow 0 \text{ Hz}$
- A.C $\rightarrow 50-60 \text{ Hz} = \infty \text{ Hz}$

$f = \infty$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{\infty} = 0$

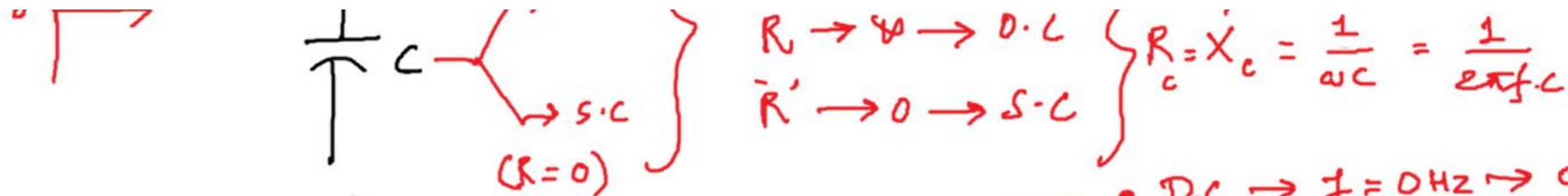
$R = 0 \rightarrow S.C$

$f = 0$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$

$R = \infty \rightarrow O.C$

$\rightarrow$ HIGH PASS } A-C Analysis.



- D.C $\rightarrow f = 0 \text{ Hz} \rightarrow 0 \text{ Hz}$
- A.C $\rightarrow 50-60 \text{ Hz} = \infty \text{ Hz}$

$t \rightarrow$
 $f = \frac{1}{t}$ ①

$f = \infty$

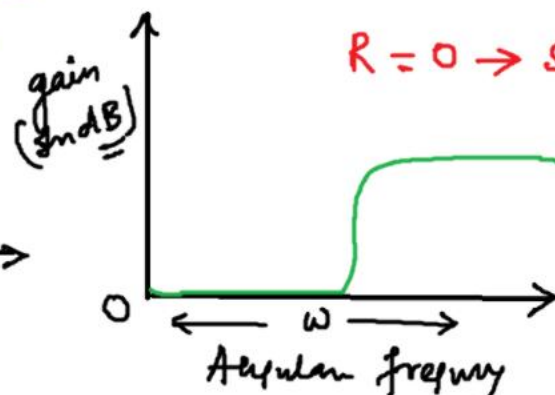
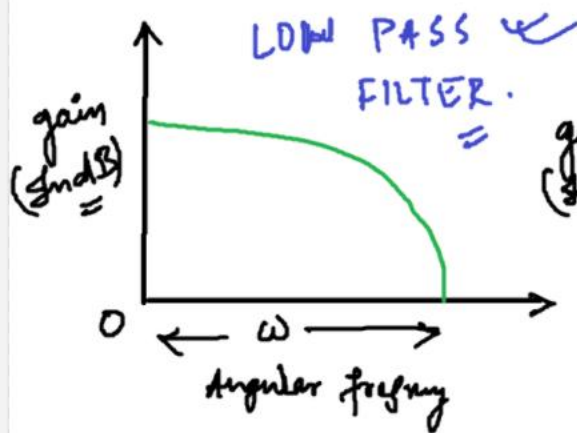
$f = 0$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{\infty} = 0$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$

$R = 0 \rightarrow \text{S.C}$

$R = \infty \rightarrow \text{O.C}$



$\rightarrow$ HIGH PASS FILTER

A-C Analysis
 AC amplitude ✓
 phase = 0° ✓